Maple Seed Funding Guide

PRODUCT AND SOFTWARE DESIGN DOCUMENT

Maple Seed Funding Guide

From the problem to a first software design: the product for business readers, and its translation into object-oriented software engineering.

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Version	0.1, working draft
Date	October 26, 2026
Status	Product design complete · software design in progress

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Contents

About this document 2

Part A. The product 3

A.1 Project overview 3

A.2 Context and motivation 3

A.3 Ideation 5

A.4 Product thinking: who it serves and what it must do 6

A.5 Main features and screens 7

A.6 Brand and identity 11

A.7 Risks, open items and roadmap 12

Part B. Software engineering 14

B.1 The software development lifecycle and where we stand 14

B.2 Requirements elicitation: actors and use cases 15

B.3 Analysis: the object model 20

B.4 Analysis: the dynamic model 23

B.5 Architectural design 25

B.6 From design to a working MVP 28

Appendix: glossary and sources 29

About this document

This document records everything decided so far about Maple Seed Funding Guide, from the original problem to a first software design, in one place.

It has two parts:

- **Part A, Product (for business readers).** What the project is, why it exists, the evidence behind it, the ideas that shaped it, what the product must do, the brand, and the risks and roadmap. No technical background is needed.
- **Part B, Software engineering (technical).** How Part A becomes software, following the object-oriented software engineering method: requirements as use cases, an analysis model in UML, and a proposed architecture. Each technical term is explained in plain words the first time it appears.

Status. The product design and a clickable prototype (English and French) are complete. The software design in Part B is a first draft, to be refined before building the MVP (minimum viable product: the smallest version that is useful). Figures in Part A come from federal sources read in October 2026 and should be rechecked before reuse.

PART A

The product

Why the project exists, the evidence behind it, what the product must do, the brand, and the roadmap. Written for business readers.

A.1 Project overview

Maple Seed Funding Guide is one place where Canadian small and medium-sized businesses (SMEs) can find, understand and prepare for the government and other support programs that fit them.

It is a bilingual website (English for Canada, French for Québec). A business owner can browse a directory of programs, answer five short questions to see which programs match their business and why, and keep a personal list that becomes a PDF report to share with an accountant or advisor. The site never applies for funding on anyone's behalf; every program links to its official website.

Item	Detail
Type	Personal, independent, non-commercial initiative
Purpose	Learning by doing: product design and software engineering, and a portfolio case study
Author	Katherine Borges Sato
Affiliation	None. Not affiliated with, endorsed by or sponsored by any government, company or educational institution
Data source	Business Benefits Finder dataset, Innovation, Science and Economic Development Canada (ISED), published under the Open Government Licence – Canada
Languages	English (Canada) and French (Québec) from the first version
Current stage	Design prototype complete; MVP not yet built

The design work lives in two companion artifacts: a design canvas with every screen in both languages, the style guide, the brand record and eleven rounds of logo exploration, and a 20-slide project presentation with speaker notes. This document turns both into prose and adds the software design.

A.2 Context and motivation

Small businesses carry a large share of Canada's economy, support for them exists at scale, yet very few use government financing and awareness of specific programs is low.

Why small and medium businesses matter

Indicator	Figure	Period
Employer businesses in Canada	1.10 million	December 2024
Share that are small (1 to 99 employees)	98.2%	December 2024
Share of private-sector jobs held by SMEs	63.6% (about 8 million people)	2024
Share of private-sector GDP from SMEs	47.2% (small 33.2% + medium 14.0%)	2022

The 47.2% is our sum of two shares the source reports separately. Anything that helps these businesses find support reaches a large part of the economy.

Where support comes from

Support for businesses comes from five kinds of providers, which the product uses as its first filter:

- **Federal government:** grants, loans, tax credits.
- **Provinces and territories:** vouchers, tax incentives.
- **Municipalities:** local programs.
- **Non-profit organizations:** financing, mentorship.
- **Private sector:** banks, accelerators.

The information is public. The federal Business Benefits Finder is even published as open data, which makes a project like this possible. Our working premise is that the hard part is not that information is missing, but finding the right program and knowing whether you qualify.

The problem, in numbers

Finding	Figure
Small businesses that requested government financing in 2024	1% (against 24% for trade credit and 9% for debt)
Government financing requests that were approved	79%
Non-users of the Canada Small Business Financing Program who knew it existed	16% (34% among its borrowers)
Those same non-users aware of any federal financing support program	34% (43% among borrowers)

Two things stand out: programs do say yes when businesses ask, and awareness of specific programs is low. The 34% and 43% figures describe awareness of federal financing programs among one program's non-users and borrowers, not among all businesses.

Reading the numbers honestly

The data supports part of the story, not all of it:

- **It supports:** programs exist at scale and approve most requests; awareness of specific programs is low (16% to 34% depending on the group); the potential audience is large (1.10 million employer businesses).
- **It does not show yet:** 79% of small businesses said they did not need financing in 2024, and only 1% named being unaware of financing sources as their main reason for not seeking it. No national figure measures how hard programs are to find or understand.

The opportunity is therefore for owners who **do** need support. The project assumes that finding and understanding programs is hard for them; it does not claim to have measured it.

A.3 Ideation

The product answers five assumed pain points with three steps, find, understand and prepare, guided by six principles.

The owner's pain: hypotheses to test

These statements come from the project's design thinking, not from a survey. They guide the design and should be checked with real business owners.

1. "I do not know which programs exist for my business."
2. "I cannot tell whether I qualify."
3. "Every program site uses its own words and steps."
4. "I do not know what to prepare before applying."
5. "A wrong guess costs time I do not have."

The proposal: find, understand, prepare

Step	Feature	What it does	Pain it answers
Find	Search by Filters	A directory of programs narrowed by provider, type of support, goal, sector, province, size and more	1, 3
Understand	Find My Support	Five short questions; programs sorted into match, maybe and not a match, with the reason for each	2, 5
Prepare	My list and PDF report	Save programs, tick off a checklist of what to prepare, and download a report to share	4, 5

Every program links to its official website. The guide never applies on behalf of the business.

Six principles behind the product

Principle	What it means in practice
Rules, not AI	Matching follows each program's published criteria. No black box.
Always explained	Every result says why: what fits, what to confirm, what does not.
Honest	No promises and no advice. Every program links to its official site.
Private by design	Nothing is saved unless the user chooses. Built with PIPEDA (the federal private-sector privacy law) and Québec's Law 25 in mind.
Bilingual	English (Canada) and French (Québec) from the first version.
Independent	A non-commercial learning project, clearly not a government website.

Privacy and legal points are design guidance, not legal advice; they need review before any public launch.

A.4 Product thinking: who it serves and what it must do

Before building software, we write down who will use it, what they are trying to get done, and what the product must and must not do. In software engineering this step is called *requirements elicitation*: drawing out and recording what the users and the client actually need. Part B.2 restates it in the formal notation engineers use.

Who uses it

User	Who they are	What they want
Visitor	An owner or manager of a Canadian SME, possibly a newcomer or early-stage founder, browsing without an account	To see quickly which programs could apply to their business
Registered user	The same person after creating a free account	To keep a list, a business profile and a report they can share
Content curator	Whoever loads program data and keeps eligibility rules current (today, the project author)	To keep the program information accurate and up to date

The third user was not visible in the screens designed so far. Writing the requirements down made it appear: someone has to maintain the program data, and the product needs tools for that.

Scenarios: the product told as short stories

- **Browsing.** Maria runs a 12-person food processing business in Ontario. She opens the site, filters by "Ontario", "Agriculture and agri-food" and "Buy equipment", reads two program pages and follows the link to an official site.
- **Guided matching.** Ahmed, a newcomer, has a two-person IT company in Québec. He answers the five questions in French, sees three matches and two "maybe" programs, each with the reasons, and downloads the checklist.
- **Preparing a meeting.** Maria creates an account, saves four programs and generates a PDF report to take to her accountant.
- **Leaving.** Ahmed decides to stop using the site and deletes all his data with one button, after a confirmation.
- **Keeping data current.** The curator imports the latest Business Benefits Finder file, reviews what changed and updates the rules of a program whose criteria changed.

What the product must do

Must have (MVP):

- Let anyone search and filter programs without an account.
- Show a program page with amount, type of support, provider, eligibility summary and a link to the official site.
- Run the five-question Find My Support flow and group results into match, maybe and not a match, with reasons per criterion.
- Show a general checklist of what to prepare.
- Let a user create an account with explicit consent, log in and log out.
- Let a registered user keep a business profile and a list of saved programs.
- Generate a PDF report of the saved list.
- Let a user download their data and delete all of it.
- Let the curator import program data and maintain each program's eligibility rules.

Should have (after the MVP): sign-in with Google, personalised recommendations on the profile page, optional owner-profile questions (for programs aimed at specific groups), email updates with opt-in.

Will not do: apply for programs, give legal, financial or tax advice, guarantee eligibility or funding, sell data or show advertising.

Qualities the product must have

These are expectations about *how well* the product works rather than *what* it does (engineers call them non-functional requirements):

- **Bilingual:** every screen, message and report in English and French.
- **Accessible:** usable with a keyboard and screen reader, aiming at WCAG 2.1 AA (the common web accessibility standard).
- **Private:** collects only what each feature needs, with separate consent for sensitive information.
- **Trustworthy:** an independence notice on every page; every program shows its source and the date it was last checked.
- **Explainable:** every match result can be traced to the rule that produced it.
- **Fast and simple:** the guided questions take about two minutes, and the site works on a phone.

A.5 Main features and screens

The prototype has nine screens, each in English and French, plus a style guide and a brand record.

Screen	What the user sees and does
Home	The promise in one line ("Funding and subsidies for Canadian SMEs, made simple"), two paths (browse or be guided), a sample PDF report and a short "how it works"
Search by Filters	A directory with 11 filter groups: offered by, type of support, main goal, amount, sector, province or territory, business size, annual revenue, business structure, owner profile, availability. Each card shows provider, type, amount and region; one click saves it
Program details	Maximum amount, type of support, provider, size limit, description, who can apply, how to apply, and a button to the official website
Find My Support	Five questions (about 2 minutes, no account), then results in three groups with reasons, a checklist, and options to download a PDF or save
About the project	The problem, the goal, the audience, MVP scope and project facts
Log in	Email and password, or Google
Create account and consent	Four steps: account, business, goals and optional owner profile, consents
Profile, list and privacy	Business profile, saved programs, PDF report, recommendations, download or delete my data
Legal notice and privacy	A plain-language notice: what the project is, what is collected and why, rights, retention, email rules and complaints

Prototype screens

The screens below come from the clickable prototype on the design canvas. They use example programs and placeholder data.

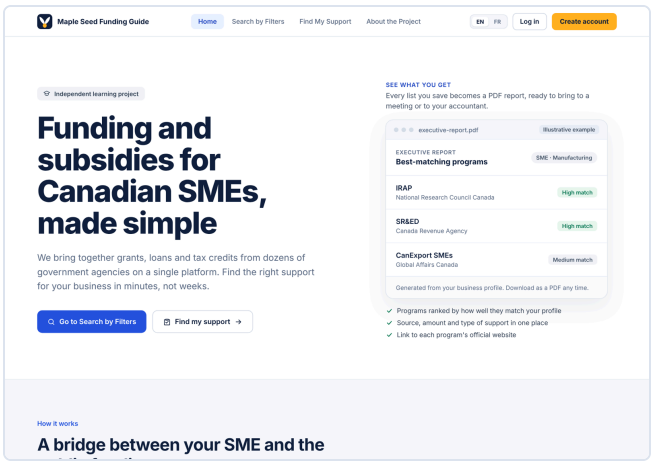


Figure 1. Home (English). The promise in one line, two paths (browse or be guided) and a sample PDF report.

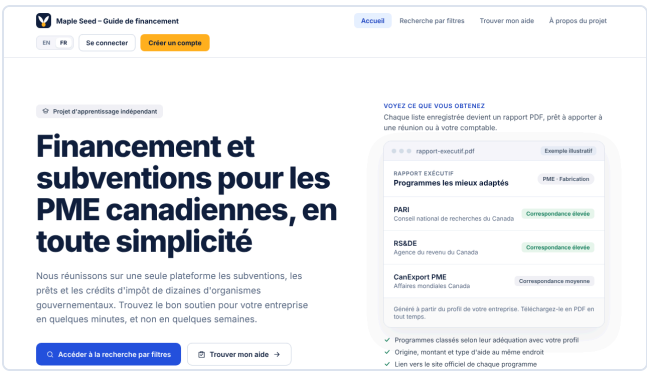


Figure 2. Home (French, Québec). Every screen exists in both languages; the descriptive part of the name is in French.

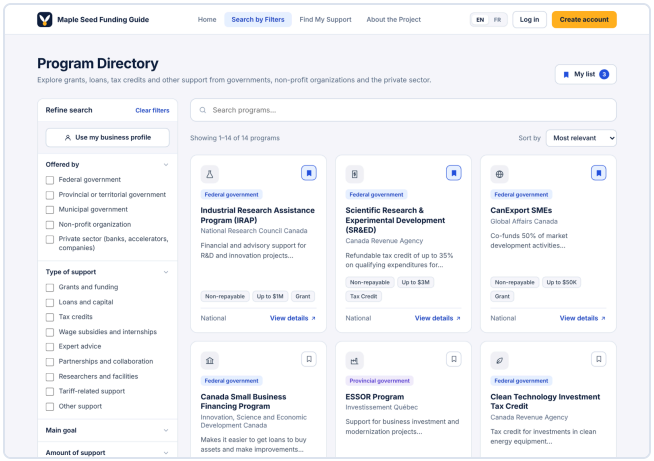


Figure 3. Search by Filters. The filter panel follows the categories of the open government data; one click saves a program.

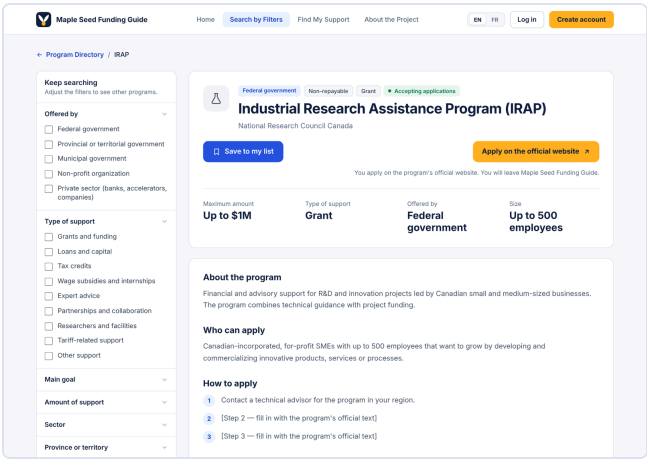


Figure 4. Program details. Amount, type of support, provider and size limit, then who can apply and a button to the official website.

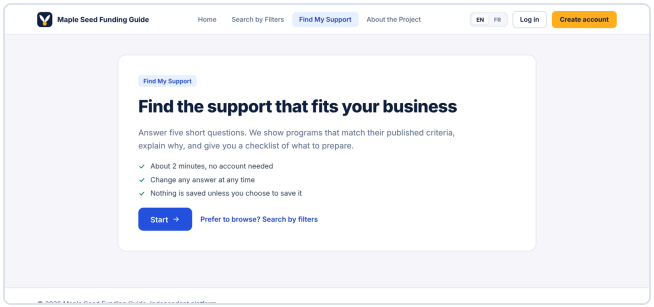


Figure 5. Find My Support, start. Five questions, about two minutes, no account needed.

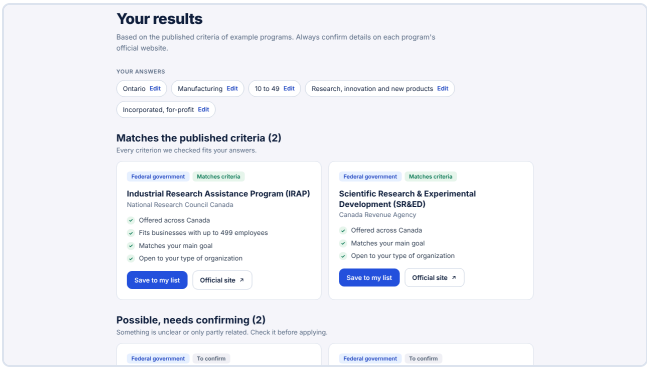


Figure 6. Find My Support, results. The user's answers, then programs grouped by match, each with the reasons per criterion.

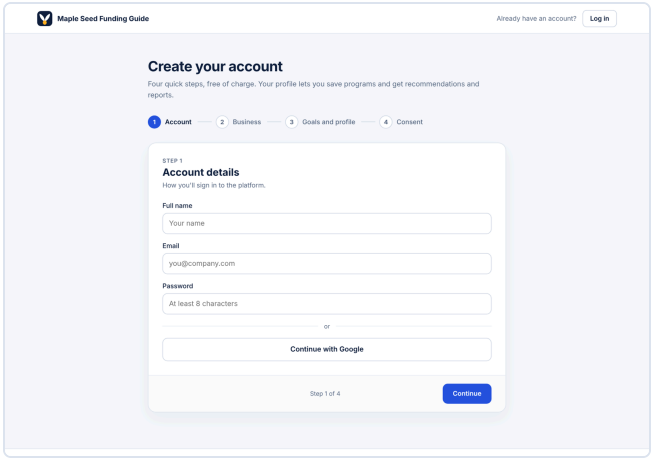


Figure 7. Create account. Four steps (account, business, goals, consent), with sensitive information kept optional.

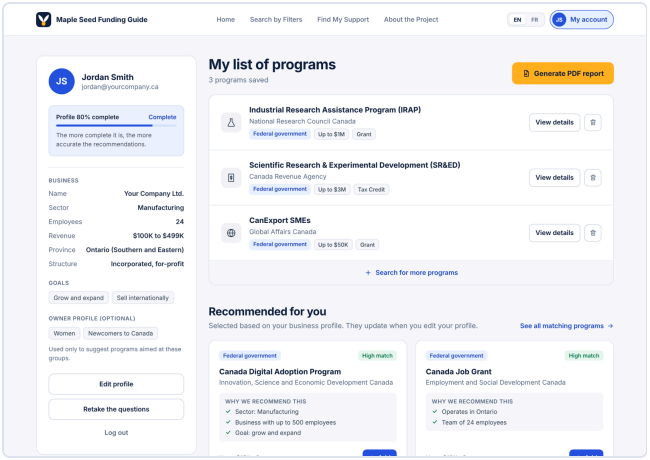


Figure 8. Profile and list. Business profile, saved programs, the PDF report and recommendations.

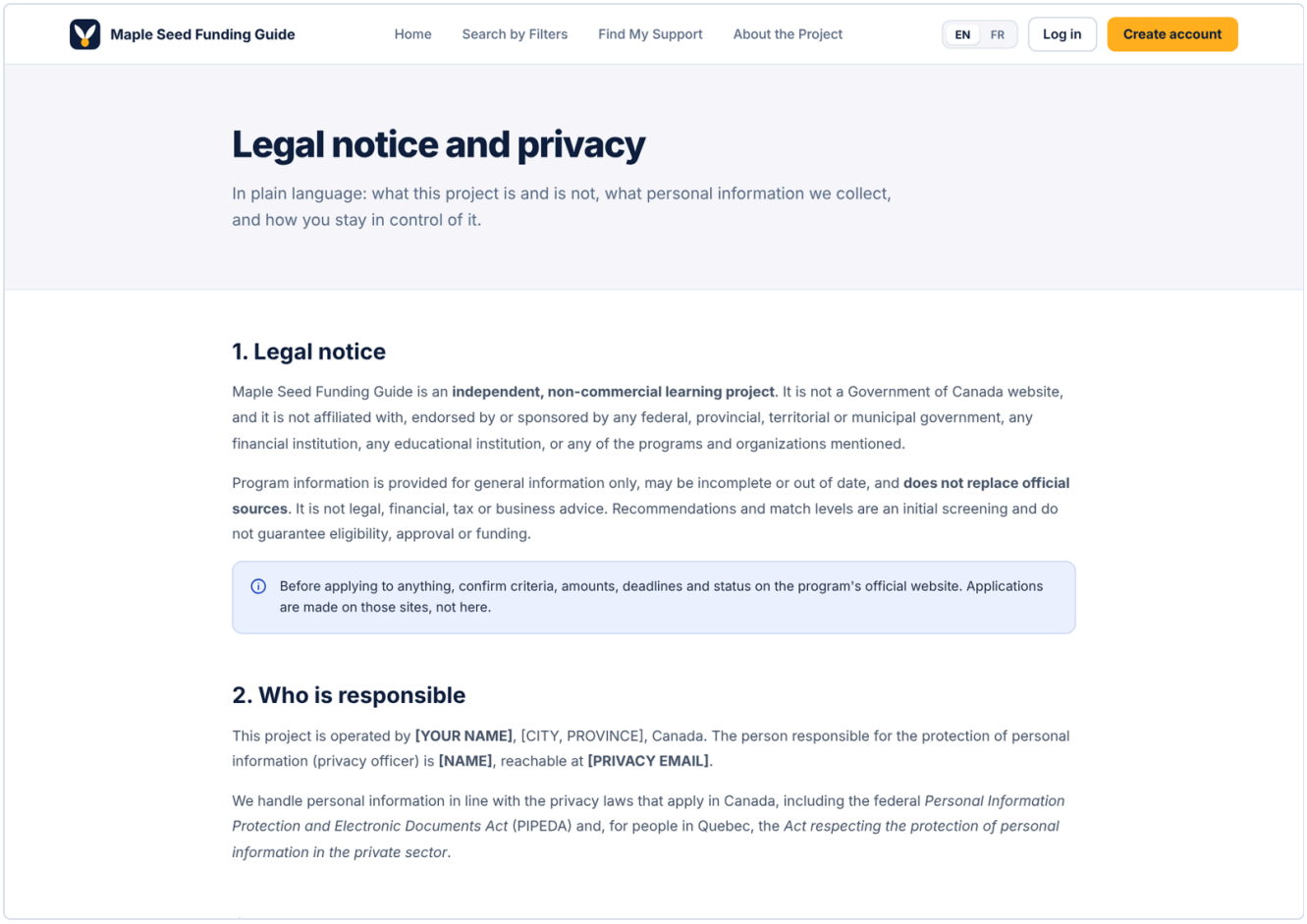


Figure 9. Legal notice and privacy, written in plain language.

The five guided questions

Question	Options	Why we ask
Where is your business based?	Ontario, Quebec, British Columbia, Alberta, another province or territory	Many programs are limited to one province or territory
What does your business do?	Manufacturing, information technology, agriculture and agri-food, clean energy, services	Some programs target specific sectors
How many employees do you have?	1 to 9, 10 to 49, 50 to 499, 500 or more	Eligibility often depends on size
What is your main goal right now?	Research and new products, sell internationally, hire or train, buy equipment or improve energy efficiency, go digital, increase working capital	Programs fund specific purposes
How is your business set up?	Incorporated for-profit, not incorporated, non-profit, not sure	Some programs are only for incorporated or non-profit organizations

How results are sorted

Each program has a short list of published criteria. For every criterion the system marks the user's answer as **fits**, **to confirm** or **does not fit**. One "does not fit" puts the program under *not a match*; only "to confirm" items put it under *maybe*; otherwise it is a *match*. The quality of the results depends entirely on how accurate and current these rules are.

Example programs in the prototype

The prototype uses 14 example programs to show the design: IRAP, SR&ED, CanExport SMEs, Canada Small Business Financing Program, ESSOR (Investissement Québec), Clean Technology Investment Tax Credit, Canada Digital Adoption Program, Alberta Innovates Micro Voucher, Canada Job Grant, CleanBC Industry Fund, Productivity and Innovation Voucher (Nova Scotia), Manitoba Small Business Venture Capital Tax Credit, Futurpreneur Canada and an example SME line of credit. Their details are simplified and not verified.

Visual style

One typeface (Inter) and colours with fixed roles: midnight blue for headings, cobalt blue for actions and links, amber for the single main call to action per screen, a light blue tint for information that is never clickable, green for confirmation, and red only for destructive actions such as deleting data. Program origin is colour-coded: blue for federal, violet for provincial, cyan for private, pink for non-profit.

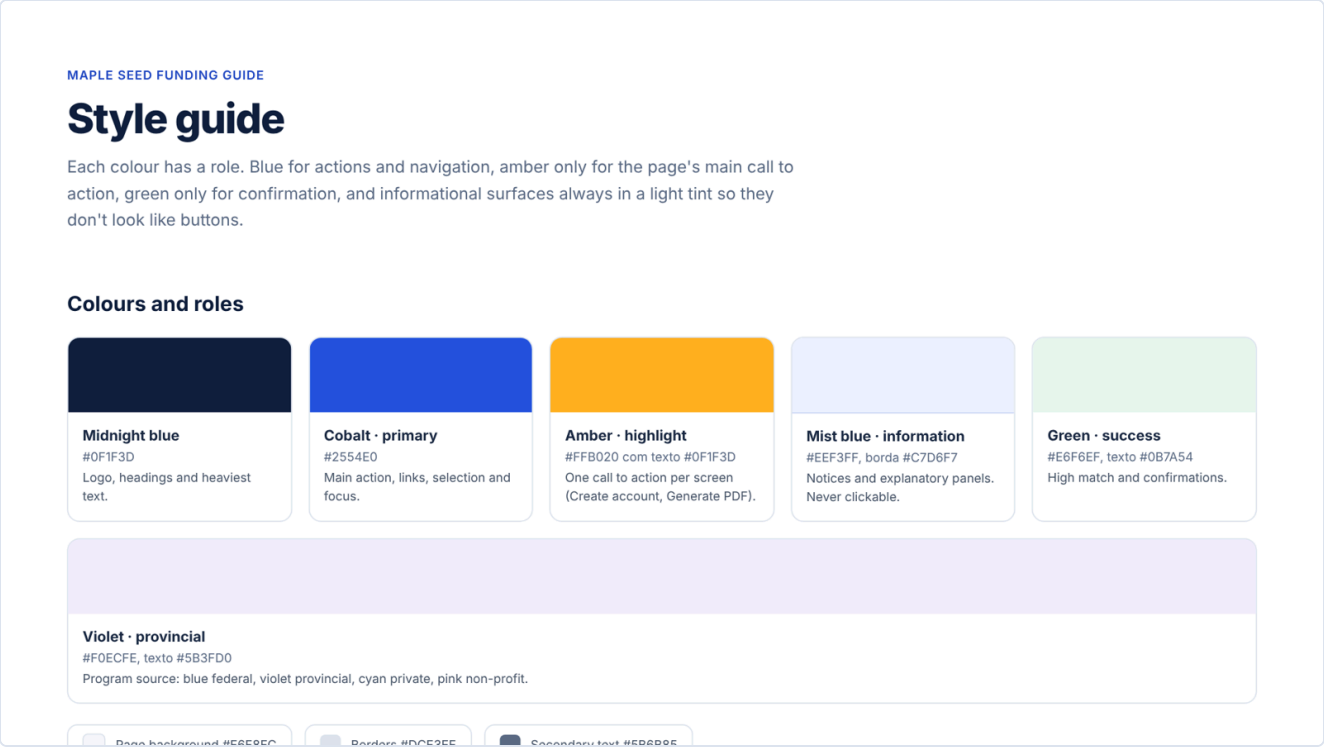


Figure 10. Style guide on the design canvas: colours and their roles, and the button hierarchy.

A.6 Brand and identity

The name is Maple Seed Funding Guide, and the symbol is a pair of maple seed wings joined at one amber seed, chosen after eleven rounds of ideas.

The name

- **Full name:** Maple Seed Funding Guide (short form: Maple Seed). In French: *Maple Seed – Guide de financement*, because Québec's Charter of the French Language asks for the descriptive part in French.
- **Meaning:** "Maple" says Canadian. "Seed" says a small start that can grow, like a young business or a first application. "Funding Guide" says what the site is without promising money.
- **What we avoided:** "Can" or "Canada" in the name, which can make a private site look like a government service, and "Fund", which suggests the site gives or manages money. Names considered along the way include CanBusiness Support, NourishCan, Samara and Maple Seed Project.

The symbol

A maple seed (a *samara*) has a flat wing and often grows in pairs joined at the seed; it spins like a helicopter when it falls. The logo shows two white wings meeting at an amber seed on a navy tile.



Figure 11. The primary logo: two white wings joined at an amber seed, on a navy tile.

Why this one:

- **Two sides, one point:** businesses and programs meeting at a match, without a literal handshake.
- **A start, not a finished thing:** fits small businesses and a learning project.
- **Canadian without the cliché:** avoids the flag's maple leaf, an official national symbol.
- **Clear when small:** one bold shape in two colours, readable at 16 pixels.
- **Sober and trustworthy:** no decoration or gradients; does not look like a bank or a government.
- **Works in one colour:** for print, stamps and PDF reports.

The logo has three versions (primary on navy, on dark backgrounds, one colour). It must not be recoloured, tilted or stretched in the header, placed next to the flag leaf or a crest, or used in a way that hides that the site is independent.

How we got there

Rounds 1 to 11 explored a guide, path and bridge; graphs and threads; handshakes; magnifiers and lanterns; clarity; plants and gardening; Canada without the cliché; NourishCan symbols; five Maple Seed ideas; a single wing; and finally refining the pair. The final mark is Round 11, refinement 1 (it began as Round 9, option 2). The full reasoning, construction grid and usage rules are in the brand and logo record on the design canvas.



Figure 12. A sample of the options explored across the eleven rounds: compass, graph, weave, magnifier, lantern, untangle, trellis, bloom, leaf, watering can, maple seed pair and single wing.

A.7 Risks, open items and roadmap

Design is done; the next phase is a small, well-tested MVP, and several checks remain before anything goes public.

Risks

Risk	How it is handled
Program data is wrong or out of date	The demo uses example data, every program links to its official page, and each program shows when it was last checked
The site looks like a government service	Independence notice on every page, no flag leaf, no "Canada" in the name
The name is already in use	Check domain, trademark and business names before going public
Handling personal data	The demo stores no real personal data; privacy is reviewed before any real use
Scope grows too large	Keep the MVP small: a few programs, done well
"Seed" reads as venture investment (Maple VC is a Canadian seed fund)	"Funding Guide" in the name; monitor reactions
The symbol reads as rabbit ears or a "V" at small sizes	Test it with people who have not seen it

Open items

- [ ] Validate the five pain hypotheses with two or three business owners (informal interviews)
- [ ] Check domains (mapleseed.ca appears taken; try mapleseedguide.ca, mapleseedfunding.ca) with a registrar or CIRA
- [ ] Search "Maple Seed" in the CIPO trademark database and the provincial business registry
- [ ] Have the French wording reviewed by a professional
- [ ] Fill in the placeholders on the About and Privacy pages: privacy officer, contact email, city, hosting location, service providers, retention period, security measures, cookie list, version date
- [ ] Complete the "How to apply" steps on program pages with official text
- [ ] Choose the technology stack (see Part B.5)
- [ ] Legal review of privacy and legal notices before any public launch

Roadmap

Phase	Content	Status
Design	Screens in English and French, guided matching prototype, brand and logo record, legal and privacy checklist	Done
Software design	Requirements as use cases, analysis model, architecture (Part B of this document)	In progress
MVP	Import program data from the Business Benefits Finder, write rules for a first set of programs, matching engine with unit tests, saved list and PDF report	Next
Polish and share	Accessibility checks (WCAG), deployed demo and README, portfolio case study, informal feedback	Then

The scope stays small on purpose: a working, well-documented MVP matters more than covering every program.

PART B

Software engineering

How the product becomes software: requirements as use cases, an analysis model in UML, and a proposed architecture.

B.1 The software development lifecycle and where we stand

The project is already in the first stages of the lifecycle: the prototype captured most of the requirements, and this part turns them into an analysis model and an architecture.

Part B follows the method taught in object-oriented software engineering (Bruegge and Dutoit, *Object-Oriented Software Engineering: Using UML, Patterns, and Java*). Software is built through seven activities, each producing a model that feeds the next:

Stage	Question it answers	Main output
Requirements elicitation	What does the client need?	Actors, use cases, non-functional requirements
Analysis	What concepts and behaviour must the system have, seen from the user's side?	Object model (class diagrams, data dictionary) and dynamic model (sequence and state diagrams)
High-level system design	How is the system split into parts, and where do they run?	Subsystems, interfaces, architectural style, deployment
Detailed object design	What exactly does each class look like in code?	Class specifications, design patterns
Implementation	Build it	Source code
Testing	Does it do what the requirements say?	Unit, integration and acceptance tests
Deployment and maintenance	Put it online and keep it working	A running service, updates

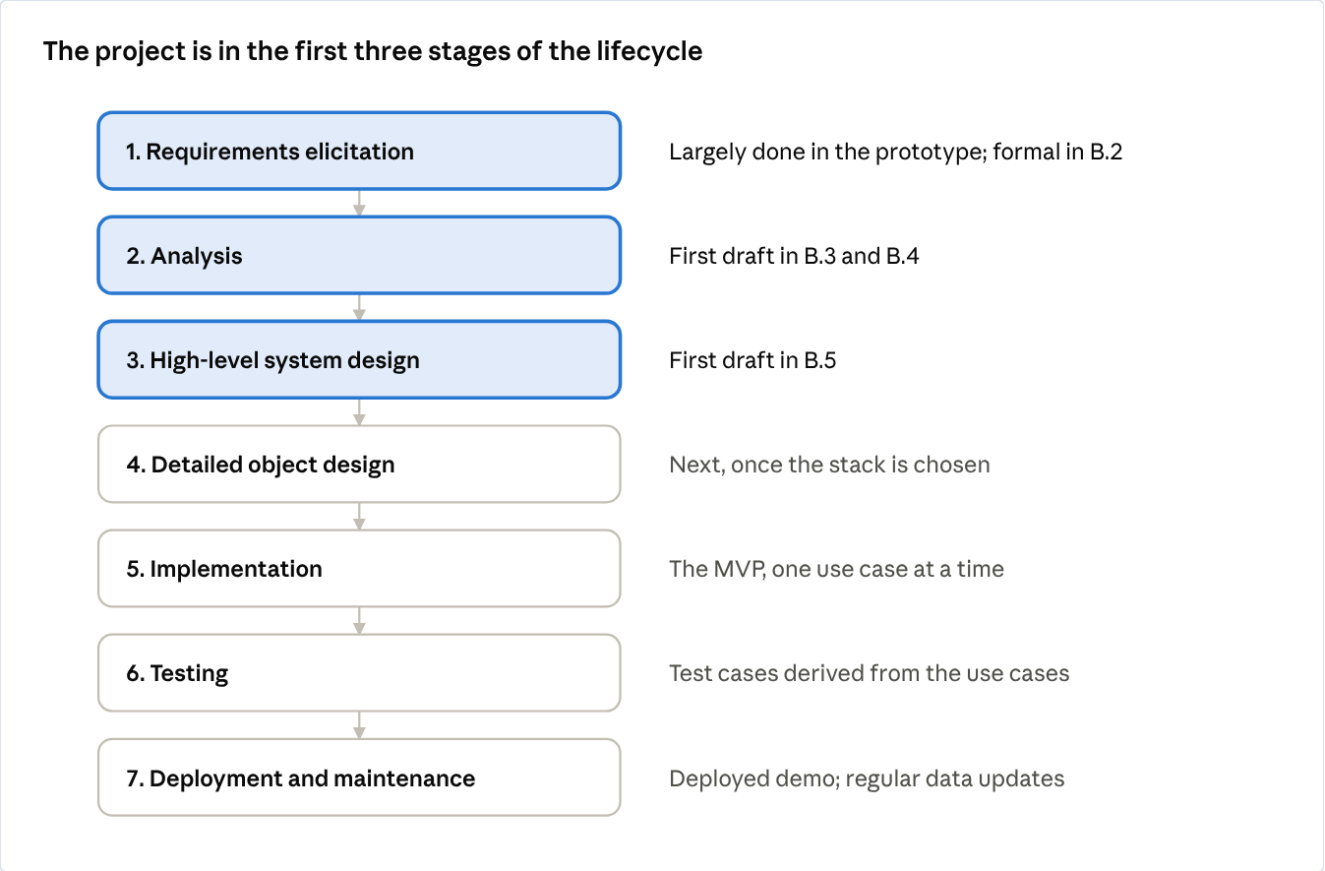


Figure 13. Software development lifecycle · 7 stages, current position highlighted

The stages are not a one-way waterfall. Each pass through them sharpens the earlier models: the clickable prototype was already a way of eliciting requirements, and writing the use cases below revealed a missing user (the content curator).

B.2 Requirements elicitation: actors and use cases

Twelve core use cases, used by three human actors and supported by external systems, describe everything the MVP must do.

A *use case* is one complete thing a user can achieve with the system, written from their point of view. An *actor* is anyone or anything outside the system that interacts with it: a type of person, or another system. The *system boundary* separates what we build from what we rely on.

Actors

Actor	Kind	Role
Visitor	End user	Anonymous SME owner; searches, reads and runs the guided questions
Registered user	End user	A visitor with an account; inherits everything a visitor can do, plus saving, reports and data controls
Content curator	End user (internal)	Loads program data and maintains eligibility rules
Business Benefits Finder dataset	External system	Source of program data (open data from ISED)
Google sign-in	External system	Optional identity provider for logging in
Email service	External system	Sends account emails (password reset, confirmations)
Official program website	External system	Where applications actually happen; the guide only links to it

System boundary

Inside the boundary: searching, explaining matches, accounts, saved lists, reports, privacy controls and data maintenance. Outside it: applying for programs, deciding eligibility (only the program can), storing program documents, and anything the official sites do. Keeping applications outside the boundary is a deliberate decision that limits legal risk and scope.

Use case diagram

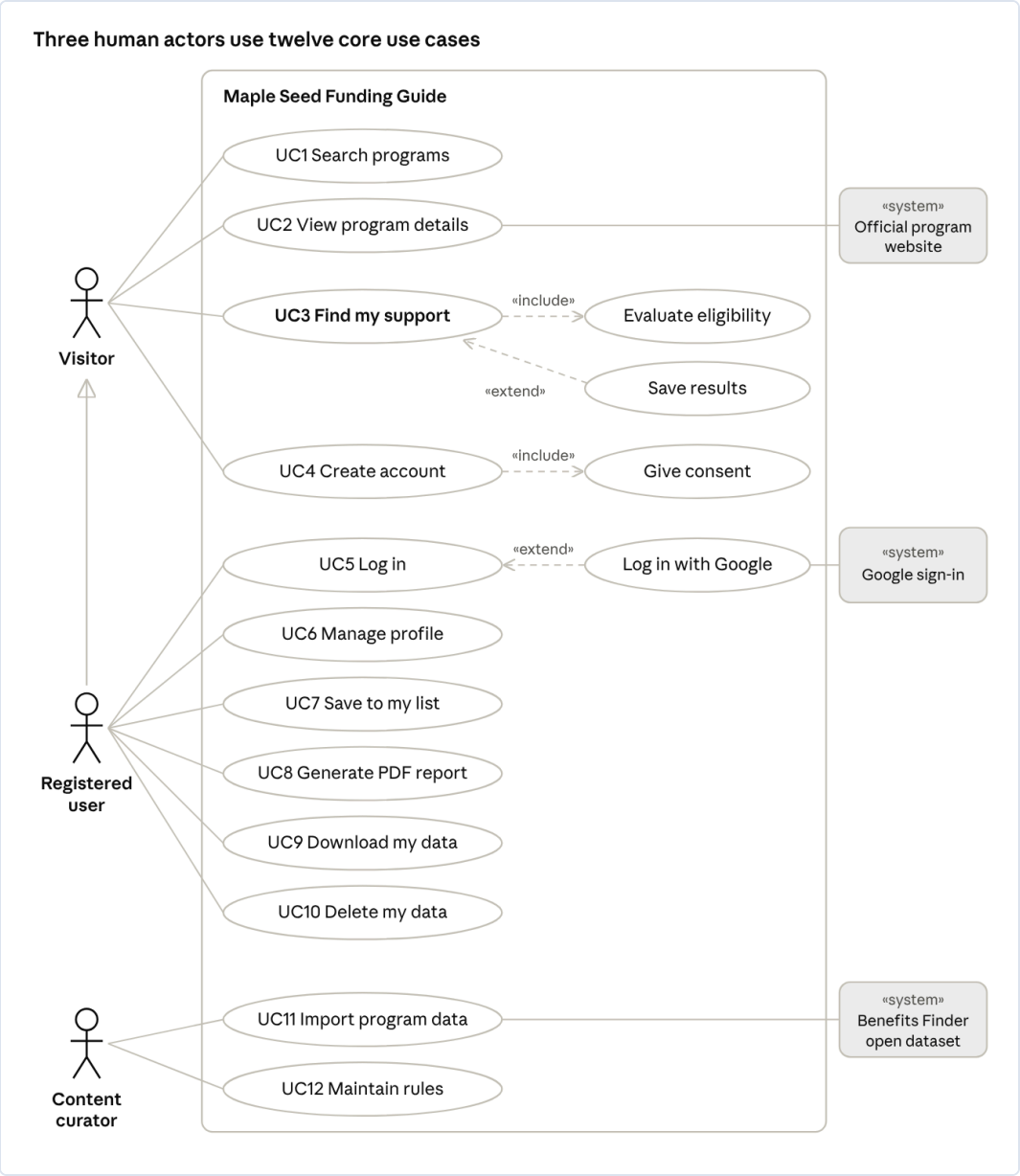


Figure 14. UML use case diagram · MVP scope

How to read it: a solid line means the actor takes part in the use case. The hollow triangle means a registered user is a kind of visitor (inheritance), so every visitor use case is also available to them. A dashed arrow marked «include» means the base use case always uses the other one; «extend» means an optional addition that happens only in some cases.

Relationships between actors and use cases

- **Initiate:** the actor starts the use case (the visitor initiates UC3 Find my support).
- **Participate:** the actor or system takes part without starting it (the Benefits Finder dataset participates in UC11; the official website participates in UC2 when the user follows the link).

Relationships between use cases

Relationship	Meaning	In Maple Seed
Include	Behaviour always reused by the base use case	UC3 Find my support includes Evaluate eligibility; UC4 Create account includes Give consent. UC1 Search also reuses Evaluate eligibility when "Use my business profile" is on
Extend	Optional behaviour added under a condition	Save results extends UC3 when the user chooses to keep them; Log in with Google extends UC5
Inheritance (generalization)	A specialised actor or use case	Registered user inherits from Visitor. UC9 and UC10 could later share a parent "Exercise a privacy right"

Use case list

ID	Use case	Initiated by	Participating	Release
UC1	Search programs	Visitor		MVP
UC2	View program details	Visitor	Official program website	MVP
UC3	Find my support	Visitor		MVP
UC4	Create account	Visitor	Email service	MVP
UC5	Log in	Registered user	Google sign-in (extension)	MVP
UC6	Manage business profile	Registered user		MVP
UC7	Save program to my list	Registered user		MVP
UC8	Generate PDF report	Registered user		MVP
UC9	Download my data	Registered user		MVP
UC10	Delete my data	Registered user		MVP
UC11	Import program data	Content curator	Business Benefits Finder dataset	MVP
UC12	Maintain eligibility rules	Content curator		MVP
UC13	Get recommendations	Registered user		Later
UC14	Subscribe to program news	Registered user	Email service	Later

Use case descriptions

Each use case gets a table with its actors, the conditions before and after, and the steps. Four key ones follow; the rest use the same template.

UC3 Find my support

Field	Content
Initiating actor	Visitor
Entry condition	The visitor is on any page and selects "Find my support"
Flow of events	1. The system shows question 1 of 5 with a short "why we ask". 2. The visitor answers; the system shows the next question. Repeat until all five are answered. 3. The system runs Evaluate eligibility (include) for every active program. 4. The system shows the answers, then programs grouped into match, maybe and not a match, each with a fits, to confirm or does not fit mark per criterion. 5. The system shows the general preparation checklist
Exit condition	The results and checklist are displayed. Nothing is stored unless Save results (extend) runs
Alternative flows	The visitor goes back to change an answer; the results recalculate. The visitor chooses "Start over"
Quality requirements	Five questions, about two minutes; works without an account; available in English and French

UC8 Generate PDF report

Field	Content
Initiating actor	Registered user
Entry condition	The user is logged in and their list holds at least one program
Flow of events	1. The user selects "Generate PDF report". 2. The system collects the business profile and the saved programs with provider, maximum amount, type of support and official link. 3. The system produces the PDF in the user's language, with the independence notice and the date. 4. The system offers the file for download
Exit condition	The user has the PDF. The list is unchanged
Alternative flow	The list is empty: the system explains how to add programs

UC10 Delete my data

Field	Content
Initiating actor	Registered user
Entry condition	The user is logged in and opens Privacy and data
Flow of events	1. The user selects "Delete my data". 2. The system explains what will be removed and that it is permanent. 3. The user confirms. 4. The system deletes the profile, list and consents, closes the account and logs the user out
Exit condition	No personal data of that user remains; a confirmation page is shown
Alternative flow	The user cancels at step 3: nothing changes

UC11 Import program data

Field	Content
Initiating actor	Content curator
Participating actor	Business Benefits Finder dataset
Entry condition	A new version of the dataset is available
Flow of events	1. The curator starts an import. 2. The system downloads the file. 3. The system cleans, normalises and validates each record. 4. The system compares it with the stored programs and lists new, changed and removed programs. 5. The curator reviews and confirms. 6. The system stores the programs and their "last checked" date; changed programs are flagged for rule review (UC12)
Exit condition	The catalogue reflects the dataset; changed programs are marked "needs review"
Alternative flow	Invalid records are reported and skipped; the import can be cancelled before step 6

Non-functional requirements

Category	Requirement
Usability	Find My Support completes in about two minutes; one main call to action per screen
Accessibility	Target WCAG 2.1 AA: keyboard use, screen-reader labels, 4.5:1 text contrast
Localisation	Every text, email and PDF in English (Canada) and French (Québec); the language can be switched on any page
Privacy and security	Minimum data per purpose; separate express consent for sensitive owner-profile data; encryption in transit; password hashing; delete and export on request (PIPEDA, Québec Law 25)
Reliability of content	Every program shows its source and last-checked date; every match result is traceable to the rule that produced it
Performance	Search and match results return in under one second for the MVP catalogue
Supportability	Rules are data, editable without changing code; core logic covered by unit tests
Legal and licensing	Attribution under the Open Government Licence – Canada; independence notice on every page; opt-in email per Canada's Anti-Spam Legislation (CASL)

B.3 Analysis: the object model

Nine core classes describe the things the system knows about; the heart of the design is a Program that owns a set of EligibilityCriteria, each able to evaluate a BusinessProfile.

The *object model* lists the concepts of the problem (classes), what each one knows (attributes), what it can do (operations), and how they relate. It is drawn as a UML class diagram. At this stage it describes the problem, not the code: names are business words, not database tables.

Three kinds of analysis objects

Kind	Purpose	Maple Seed examples
Entity objects	Long-lived information the system keeps	Program, Provider, EligibilityCriterion, User, BusinessProfile, Consent, SavedList
Boundary objects	Where actors interact with the system	SearchPage, ProgramPage, QuestionnaireForm, ResultsPage, PdfReport, ImportScreen, BenefitsFinderGateway
Control objects	Coordinate one use case	MatchingController (UC3), ReportController (UC8), ImportController (UC11), AccountController (UC4, UC10)

The diagram below shows the entity objects, which carry the domain knowledge.

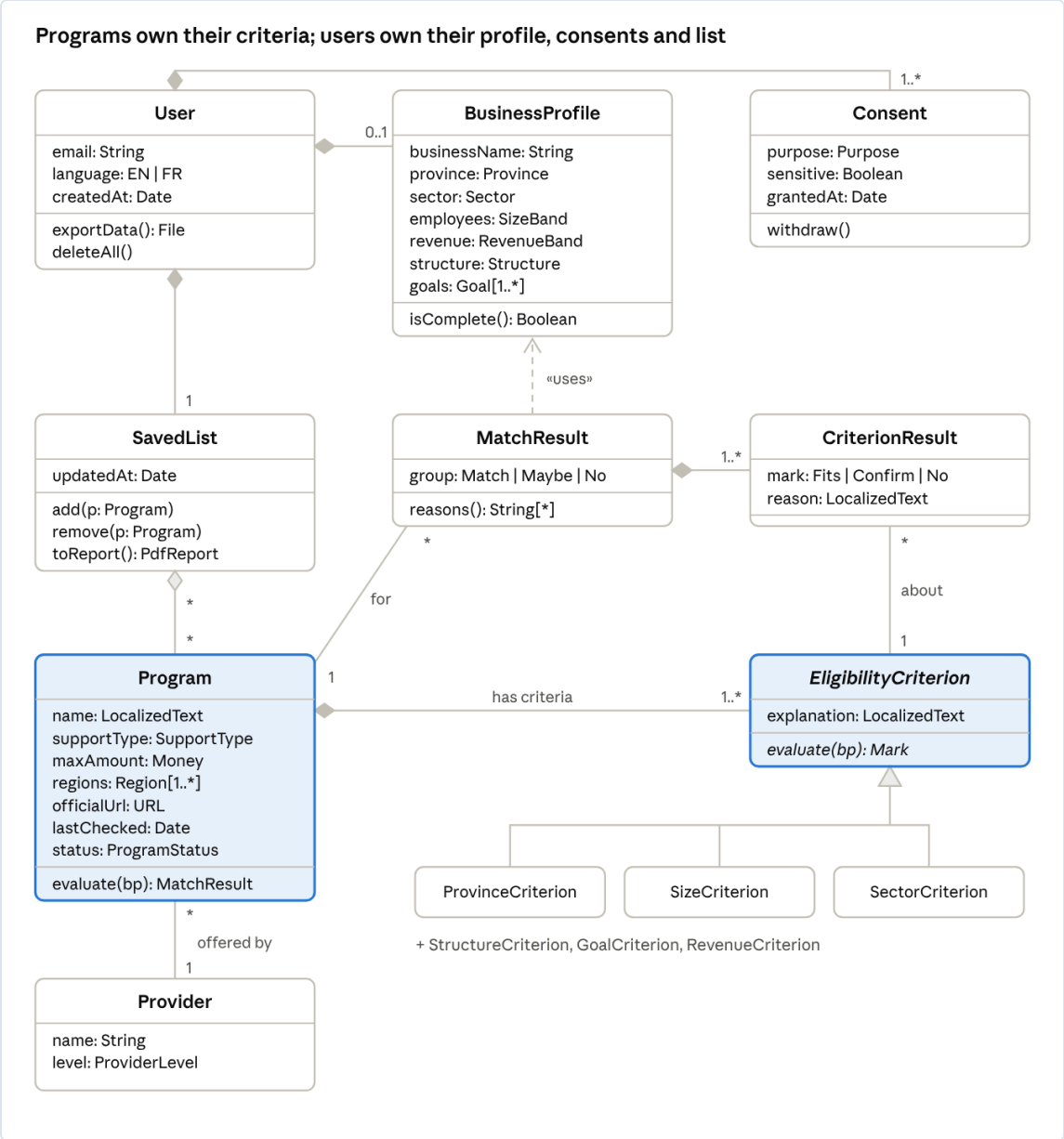


Figure 15. UML class diagram · analysis object model, entity classes

How to read the relationships

Notation	Meaning	Where it appears
Filled diamond (composition)	The part cannot exist without its whole and is deleted with it	A Program's criteria; a User's profile, consents and list; a MatchResult's criterion results
Hollow diamond (shared aggregation)	The whole groups parts that also live on their own	A SavedList holds Programs, but a Program exists regardless of any list, and many lists can hold the same Program
Hollow triangle (inheritance)	A specialised class is a kind of the general one	ProvinceCriterion, SizeCriterion, SectorCriterion and others are kinds of EligibilityCriterion
Numbers at line ends (multiplicity)	How many objects take part	A User has 0..1 BusinessProfile; a Program has 1..* criteria; a Provider offers * (many) programs
Dashed arrow	One class depends on another without owning it	A MatchResult is computed using a BusinessProfile (or the anonymous answers, which have the same shape)

Directionality. Associations are navigable in the direction the use cases need: a SavedList knows its Programs, but a Program does not need to know which lists contain it. A CriterionResult points to the criterion it explains. Keeping links one-way lowers coupling.

Why inheritance for criteria

Each type of criterion checks a profile differently: a province criterion compares regions, a size criterion compares employee bands, a goal criterion looks for overlap between goals. Making them subclasses of an abstract EligibilityCriterion, each with its own *evaluate(profile)*, lets the matching engine treat them all the same way (polymorphism) and lets a new type be added without changing the engine. Rules stay data: the curator sets the values ("Ontario only", "up to 500 employees"), and the subclass supplies the logic.

Data dictionary

Term	Definition
Program	A source of support for businesses (grant, loan, tax credit, voucher, advice) offered by one Provider, with an official website where applications are made
Provider	The organization that offers a program. ProviderLevel is one of federal, provincial or territorial, municipal, non-profit, private
SupportType	Grant, loan, tax credit, voucher, wage subsidy, advisory service, equity or other
Region	A province or territory, or "all of Canada"
ProgramStatus	Active (accepting applications), Closed, or Needs review (data changed since the rules were last checked)
EligibilityCriterion	One published condition of a program, such as "based in Ontario" or "up to 500 employees", with a plain-language explanation in both languages
Mark	The outcome of checking one criterion: Fits, To confirm (unclear or only partly related), or Does not fit
CriterionResult	One Mark plus the reason shown to the user
MatchResult	The outcome for one program and one profile: Match (all criteria fit), Maybe (no criterion fails, at least one to confirm) or No (at least one does not fit)
BusinessProfile	The business facts used for matching: province, sector, size band, revenue band, structure and goals
SizeBand	1 to 9, 10 to 49, 50 to 499, 500 or more employees

Term	Definition
Consent	A recorded permission for one purpose (account, recommendations, sensitive owner profile, email), with its date; it can be withdrawn
SavedList	The programs a registered user has kept; the source of the PDF report
LocalizedText	A text held in both English and French
PdfReport	A generated document with the profile, saved programs, sources, the date and the independence notice

B.4 Analysis: the dynamic model

The dynamic model shows behaviour over time: who sends which message to whom during a use case, and which states an object passes through.

The object model says what exists; the dynamic model says what happens. It is drawn from the user's perspective, one use case at a time, and it often reveals operations missing from the class diagram.

Sequence diagram: UC3 Find my support

A sequence diagram reads top to bottom as time passes. Each column is an object (its *lifeline*); each horizontal arrow is a message. Solid arrows are requests, dashed arrows are replies, and boxes marked *loop* repeat.

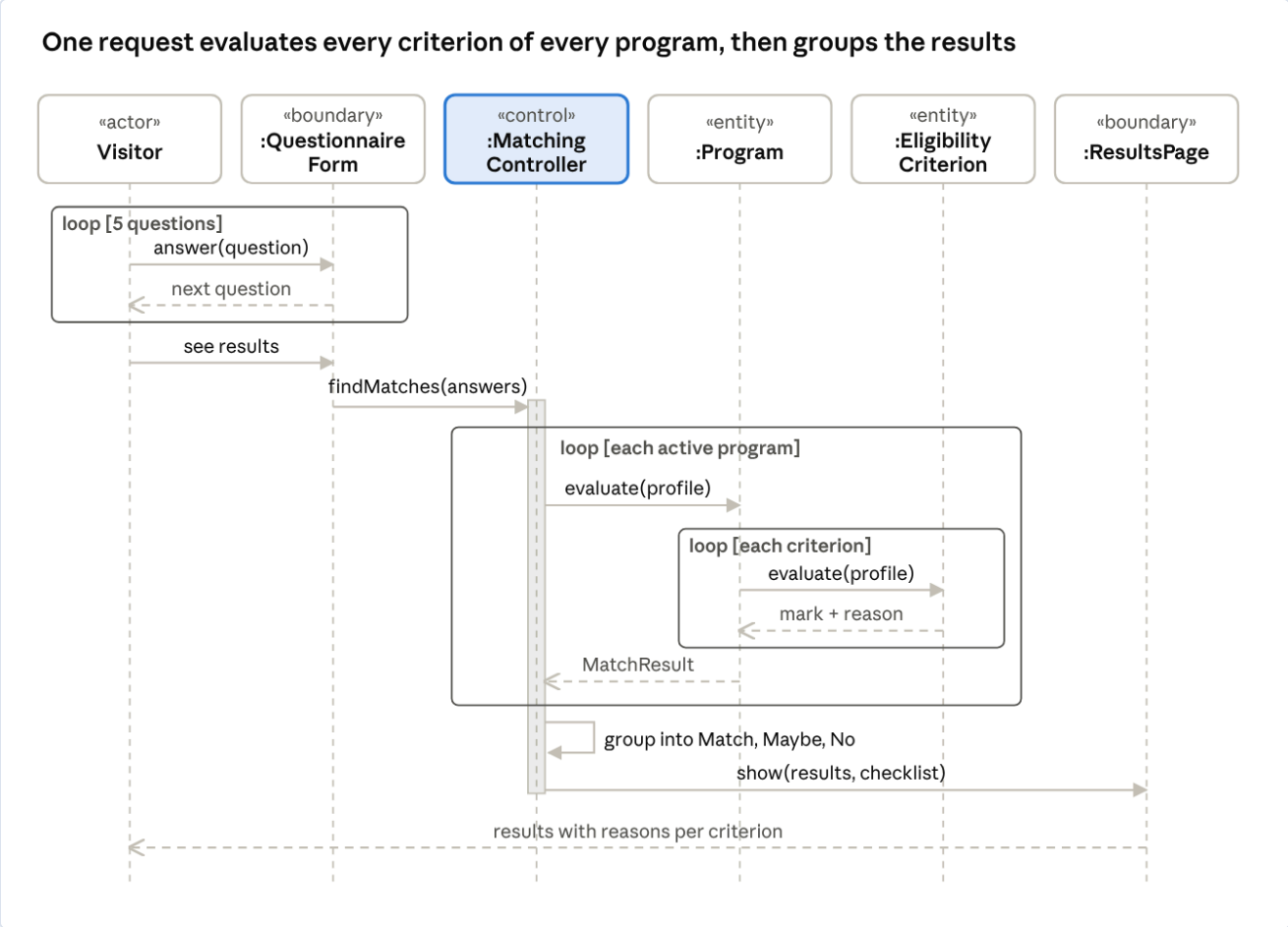


Figure 16. UML sequence diagram · UC3 Find my support

What it shows in plain words: the visitor answers five questions through the form. When they ask for results, the form hands the answers to the matching controller, which asks each active program to evaluate the profile. Each program asks each of its criteria for a mark and a reason, and returns a MatchResult. The controller groups the results into Match, Maybe and No and hands them, with the checklist, to the results page.

What it teaches the design: the controller never checks a rule itself. Programs and criteria do, which keeps the logic in one place and easy to test. The same evaluate operation serves UC1 ("Use my business profile" in search) and UC13 (recommendations).

State machine diagrams

A state machine shows the states one object can be in and the events that move it between them. Labels read *event [condition] / action*.

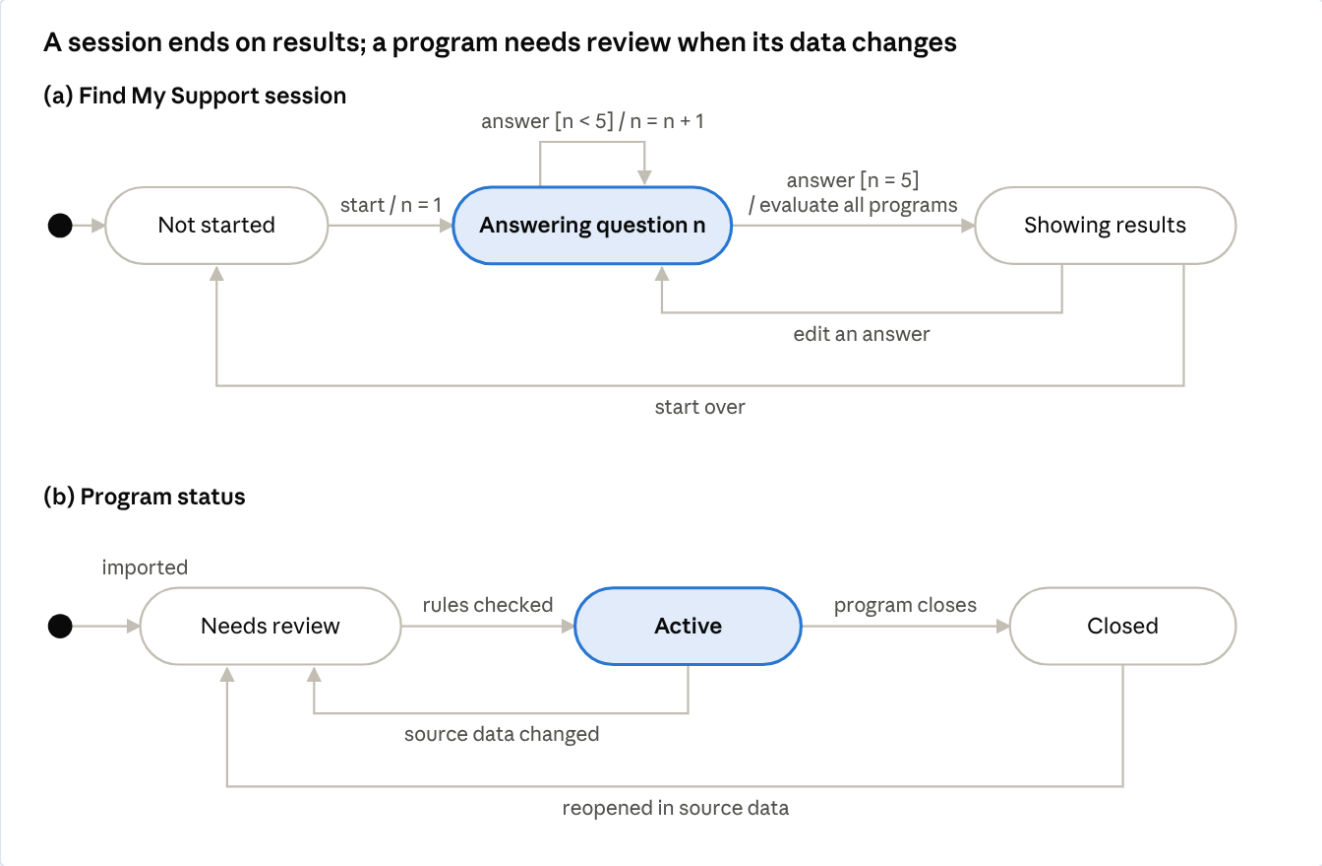


Figure 17. UML state machine diagrams · questionnaire session and program status

- **(a) Find My Support session.** The visitor moves from question to question until the fifth answer triggers the evaluation. From the results they can edit any answer (which returns to that question and recalculates) or start over. Nothing is saved unless they choose Save results.
- **(b) Program status.** Every imported or changed program starts in *Needs review*, and only a curator's check of its rules makes it *Active*. This is how the design keeps the honesty principle: a program whose source changed is never matched on stale rules without being flagged.

A third state machine worth drawing during detailed design is the **account lifecycle**: Guest, then Registered (after consent), then Deleted (after UC10), with consents that can be withdrawn individually while the account stays active.

B.5 Architectural design

The proposal is a three-tier web application, built in the Model-View-Controller style, with five subsystems, two separate data stores and a pipe-and-filter import pipeline.

Architectural design (also called system design) decides how the system is split into parts, how those parts talk to each other, and where they run. Good choices here make the rest of the work easier; poor ones are expensive to undo.

Key terms

Term	Meaning	Example here
Class	The smallest design unit: data plus behaviour	Program, EligibilityCriterion
Subsystem	A group of related classes with one clear responsibility	Matching (the rule engine and its classes)
Component	A replaceable, deployable piece of software that provides a subsystem's functionality	The matching library, the PDF generator
Service	A set of related operations a subsystem offers to others through its interface	evaluate(profile, programs) offered by Matching
Subsystem interface	The only door into a subsystem; others call it and never reach inside	Catalogue offers search(filters) and getProgram(id)
Node	A physical or virtual machine where components run	The user's browser, the application server, the database server

Proposed architecture

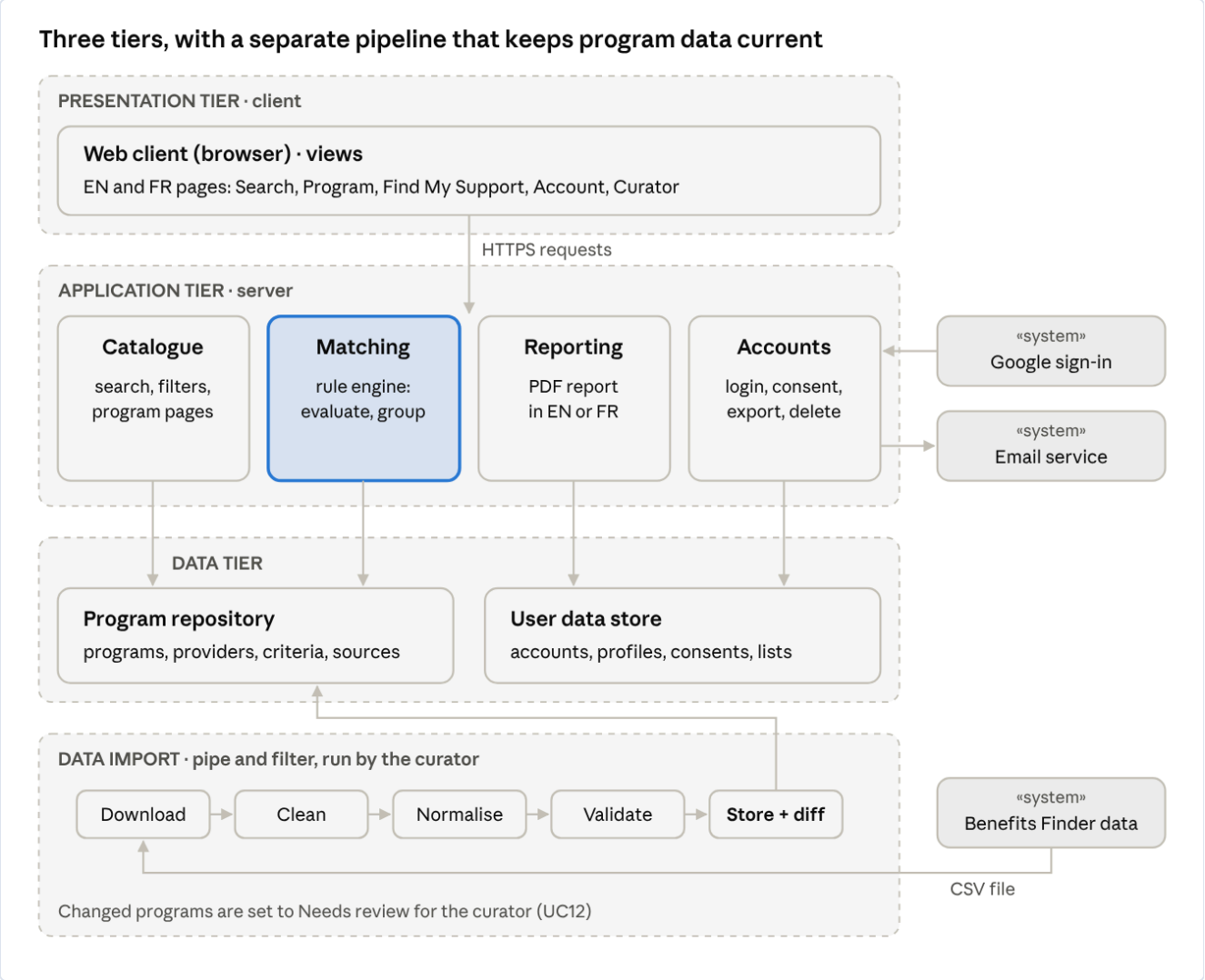


Figure 18. Proposed architecture · tiers, subsystems, repositories and external systems

Requests travel down from the browser to the application tier, which reads and writes the data tier. The import pipeline is separate: it is run by the curator, not by visitors, and it only writes to the program repository.

Subsystems and their interfaces

Subsystem	Responsibility	Interface (services offered)	Use cases
Catalogue	Programs, providers, filters and program pages	search(filters, language), getProgram(id)	UC1, UC2
Matching	Evaluate a profile against program rules and group the results	evaluate(profile): MatchResult[*], checklist()	UC3, UC13
Reporting	Produce the PDF report in the user's language	generateReport(user): PdfReport	UC8
Accounts and privacy	Sign-up, consent, login, profile, list, data export and deletion	register, login, saveProgram, exportData, deleteAll	UC4 to UC7, UC9, UC10
Data import and curation	Bring in the open dataset and maintain rules	runImport(file), listChanges(), updateRules(program)	UC11, UC12

Coupling and cohesion

- **High cohesion:** each subsystem does one job. Matching knows nothing about web pages, PDFs or accounts; it takes a profile and programs and returns results. That makes it the easiest part to unit test and to explain in a portfolio.
- **Low coupling:** subsystems talk only through their interfaces. Reporting asks Catalogue for program details rather than reading its tables directly, so the database can change without breaking reports.
- **Privacy by separation:** personal data lives in its own store, apart from public program data. Deleting a user touches only the user data store, and the program repository holds no personal information at all.

Architectural styles considered

Style	What it is	Fit for Maple Seed
N-tier (three-tier)	Presentation, application and data layers, each calling only the one below	**Chosen** as the overall structure
Model-View-Controller (MVC)	Views show data, models hold it, controllers handle user actions and connect them	**Chosen** inside the application: pages are views, entity classes are models, the control objects from B.3 are controllers
Client-server	Many clients send requests to one server	**Chosen**: browsers are clients of one web application
Repository	Subsystems share one central data store	**Chosen** for program data: Catalogue, Matching and Reporting all read the program repository
Pipe and filter	Data flows through a chain of independent steps, each transforming it	**Chosen** for data import: download, clean, normalise, validate, store; each step testable on its own
Peer-to-peer	Nodes act as both client and server with no central server	Not suitable: we need one trusted, curated source of program data

Deployment (nodes)

Node	What runs there
User's device	A web browser showing the views; no installation
Application server	The web application: controllers and the five subsystems
Database server	The program repository and the user data store, ideally in two separate databases or schemas
External services	Google sign-in, an email service, the open data portal

Hosting in Canada is preferred, so personal data does not leave the country (this also simplifies the privacy notice, which must name any transfer outside Québec or Canada).

Technology stack (open decision)

The architecture does not depend on a specific stack. Any mainstream web framework with a relational database fits. Criteria for choosing: a language already being learned in the program, good testing tools, built-in support for translations (EN and FR) and simple, low-cost hosting in Canada. The choice should be recorded here once made.

B.6 From design to a working MVP

Build the MVP one use case at a time, with each use case's entry and exit conditions turned into tests, so every requirement can be traced to code and back.

Next engineering steps

1. **Refine the models.** Review the use case diagram and class diagram above, add the account lifecycle state machine, and draw sequence diagrams for UC8 (PDF report) and UC11 (import).
2. **Choose the stack** and record it in B.5.
3. **Detailed object design.** Turn analysis classes into code-level classes: exact types, visibility, and design patterns where they help. Candidates: *Strategy* for criterion types (already implied by the inheritance in B.3), *Facade* for each subsystem interface, *Adapter* for the Benefits Finder file format, *Template Method* for the import pipeline steps.
4. **Implement in thin vertical slices**, each a working feature from page to database:
    1. UC11 Import program data (small sample) and UC1 Search programs
    2. UC2 View program details
    3. UC12 Maintain rules, then UC3 Find my support with the matching engine
    4. UC4 Create account and UC5 Log in
    5. UC6 Manage profile, UC7 Save to my list, UC8 Generate PDF report
    6. UC9 Download my data and UC10 Delete my data

Polish: accessibility checks against WCAG 2.1 AA, French review, README and deployed demo.

Testing and traceability

Each requirement should be traceable through the models to code and tests. A test is written from each use case's entry condition, flow and exit condition.

Requirement	Use case	Subsystem and classes	Example test
Show why a program matches	UC3	Matching: Program, EligibilityCriterion, MatchResult	A profile in Ontario against an "Ontario only" criterion returns Fits with the reason text
Group results correctly	UC3	Matching: MatchResult	One Does not fit gives No; only To confirm gives Maybe; all Fits gives Match
Delete all personal data	UC10	Accounts: User, BusinessProfile, Consent, SavedList	After deleteAll(), no record of that user remains in the user data store
Flag changed programs	UC11	Data import: ImportController	A changed program in the new file ends in Needs review
Bilingual report	UC8	Reporting: PdfReport	A report requested in French has every heading and program name in French

Test levels: **unit tests** for each criterion type and the grouping rule; **integration tests** for each subsystem interface; **acceptance tests** that walk through each use case as a user would.

Deployment and maintenance

- **Configuration management:** version control from the first line of code, with small, described commits.
- **Content maintenance:** a regular import of the Benefits Finder dataset (for example, monthly), with every changed program reviewed before it is matched again.
- **Monitoring the promise:** the "last checked" date on each program shows visitors how current the information is.

Appendix: glossary and sources

Glossary

Term	Meaning
SME	Small and medium-sized enterprise. In the statistics used here, small = 1 to 99 employees, medium = 100 to 499
ISED	Innovation, Science and Economic Development Canada, the federal department that publishes the Business Benefits Finder and the small business statistics
MVP	Minimum viable product: the smallest version that is useful and can be tested with users
UML	Unified Modeling Language, the standard notation for software models (use case, class, sequence and state diagrams)
PIPEDA	Personal Information Protection and Electronic Documents Act, the federal privacy law for the private sector
Law 25	Québec's modernised Act respecting the protection of personal information in the private sector
CASL	Canada's Anti-Spam Legislation, which requires consent before sending commercial email
WCAG	Web Content Accessibility Guidelines; level AA is the common target
CIPO	Canadian Intellectual Property Office, where trademarks are searched and registered
CIRA	Canadian Internet Registration Authority, which manages .ca domains
IRAP, SR&ED	Two federal programs used as examples: the Industrial Research Assistance Program and the Scientific Research and Experimental Development tax incentive

Sources

The statistics in Part A were read from these federal publications in October 2026 and should be rechecked before reuse:

- Innovation, Science and Economic Development Canada, *Key Small Business Statistics 2025*. Business counts to December 2024, employment 2024, GDP shares 2022.
- Innovation, Science and Economic Development Canada, *Credit Conditions Survey 2024*. 1,825 small businesses surveyed January to April 2025, covering financing sought in 2024.
- Innovation, Science and Economic Development Canada, *Borrower Awareness and Satisfaction Study 2023–2024*, Canada Small Business Financing Program. Fieldwork November 2023 to January 2024.
- *Business Benefits Finder* dataset, Open Government portal, published under the Open Government Licence – Canada.

Method reference: Bernd Bruegge and Allen H. Dutoit, *Object-Oriented Software Engineering: Using UML, Patterns, and Java*, 3rd edition, Pearson, 2009.

Companion artifacts: the Maple Seed Funding Guide design canvas (screens, style guide, brand record) and the project presentation (20 slides).